Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.23	20.10	
worker nodes in service, mean	6.00	6.00	+0.0%
worker nodes in service, min	6.00	6.00	+0.0%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.02	2.01	-0.7%
power (W), mean	629	643	+2.1%
power (W), peak	651	687	+5.5%
energy (Wh)	212	215	+1.4%
energy, parked workers still on at idle power (Wh)	212	215	+1.4%
CPU used (cores), mean	0.901	1.25	+39.2%
CPU allocatable (cores), mean	24.00	24.00	+0.0%
utilisation (used / allocatable)	0.038	0.052	+39.2%
energy per core-hour (Wh)	698	512	-26.6%
node-hours per core-hour	6.66	4.78	-28.2%
pending pods, pod-minutes	0.533	0.000	-100.0%
pending pods, peak	2.00	0.000	-100.0%
HPA replicas, mean	9.07	9.42	+3.9%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.533	0.533	+0.0%
requests timed	1,020	1,095	+7.4%
time over the response line (% of samples)	0.882	0.000	-100.0%
response time (ms), mean	179	97.02	-45.7%
response time (ms), median	144	82.30	-42.9%
response time (ms), 95th percentile	357	158	-55.6%
response time (ms), 99th percentile	494	205	-58.5%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 9; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	7.2705	5.2131	-28.3%	[-3.6429, -0.4448]	GAIN
kwh_per_core_hour	0.7590	0.5548	-26.9%	[-0.3634, -0.0464]	GAIN
utilisation	0.0371	0.0533	+43.7%	[+0.0032, +0.0304]	GAIN
pending_pod_minutes_per_hour	1.5118	0.0000	-100.0%	[-4.5354, +0.0000]	STEADY, no significant difference
hpa_shortfall_minutes_per_hour	1.5178	1.6865	+11.1%	[-2.9450, +3.3662]	STEADY, no significant difference

What Omni-Compass did

- decisions: 20 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 0
- power cap decided per minute: 1.0 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695
- heat (thermal) per minute: 0.358 0.392 0.421 0.445 0.468 0.488 0.505 0.521 0.534 0.547 0.557 0.563 0.568 0.572 0.577 0.581 0.585 0.588 0.59 0.591
- nodes decided per minute: 6 5 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	618	620	5.00	6.00	0.000	0.000	0.575	0.663

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	6.00	617	627	6.00	7.00	0.000	0.000	0.582	0.821
2	6.00	6.00	619	624	6.00	8.00	0.000	0.000	0.636	0.776
3	6.00	6.00	622	632	6.00	8.00	0.000	0.000	0.699	0.977
4	6.00	6.00	633	651	9.00	10.00	0.500	0.000	1.00	1.48
5	6.00	6.00	634	650	10.00	10.00	0.000	0.000	1.00	1.44
6	6.00	6.00	634	651	10.00	10.00	0.000	0.000	1.00	1.46
7	6.00	6.00	647	672	10.00	10.00	0.000	0.000	1.37	2.06
8	6.00	6.00	647	673	10.00	10.00	0.000	0.000	1.37	2.06
9	6.00	6.00	648	679	10.00	10.00	0.000	0.000	1.39	2.17
10	6.00	6.00	638	664	10.00	10.00	0.000	0.000	1.39	1.80
11	6.00	6.00	620	625	10.00	10.00	0.000	0.000	0.639	0.784
12	6.00	6.00	618	623	10.00	10.00	0.000	0.000	0.627	0.763
13	6.00	6.00	624	632	10.00	10.00	0.000	0.000	0.770	0.996
14	6.00	6.00	633	650	10.00	10.00	0.000	0.000	1.00	1.43
15	6.00	6.00	637	647	10.00	10.00	0.000	0.000	1.06	1.38
16	6.00	6.00	634	646	10.00	10.00	0.000	0.000	0.996	1.36
17	6.00	6.00	627	640	10.00	10.00	0.000	0.000	0.844	1.22
18	6.00	6.00	620	626	10.00	10.00	0.000	0.000	0.621	0.788
19	6.00	6.00	618	624	10.00	10.00	0.000	0.000	0.615	0.755

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.